Minimum-Repair Matroids and a Basis-Exchange Atlas for a Frozen Hénon Core

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Abstract

For each deletion set in a frozen sixteen-label finite model, we classify the entire family of minimum restorations that recover the full core. Every such family is the set of bases of an explicit direct sum: irrelevant deleted labels are loops, the deleted pivot is a coloop, and the remaining component is a rank-truncated partition matroid on the fully deleted direction blocks. Exhaustive point-set closure enumeration verifies this identification and basis exchange for all 65536 deletion sets. The ten repair/multiplicity templates yield exactly five unlabeled exchange graphs, K_1, K_4, K_7, K_8, and $L(K_{1,1,2,5})$. In the all-deleted case the 25 bases equal the 25 minimal triples of the predecessor closure atlas. This is a finite combinatorial theorem under the firewall `NO_BAD_EULER_OR_ROOT_NUMBER`.

1 Frozen repair problem

Let $L = \{S_1, \dots, S_{16}\}$ and let $D \subseteq L$ be deleted, with $A = L \setminus D$ retained. The pivot is $p = S_9$, and the four direction blocks are

$$B_1 = \{S_1\}, \quad B_2 = \{S_{16}\}, \quad B_3 = \{S_7, S_{15}\}, \quad B_4 = \{S_3, S_4, S_8, S_{11}, S_{12}\}.$$

The other six labels are dummy for full-core generation. Write

$$I(D) = \{i : B_i \subseteq D\}, \qquad t(D) = |I(D)|, \qquad r(D) = \max\{0, t(D) - 2\}.$$

A restoration witness is $R \subseteq D$ for which $A \cup R$ generates the full finite core, and it is minimum when $|R|$ is least possible.

2 Matroid theorem

Let

$$E_D = D \setminus \left((\{p\} \cap D) \cup \bigcup_{i \in I(D)} B_i \right)$$

and let $P_D = \bigoplus_{i \in I(D)} U_{1,B_i}$ be the partition matroid with capacity one in each fully deleted block. Define

$$M_D = U_{0,E_D} \oplus U_{1,\{p\} \cap D} \oplus \mathrm{Tr}_{r(D)}(P_D),$$

where the middle summand is absent when $p \notin D$.

Theorem 1 (Minimum-repair basis identification)**.** *For every $D \subseteq L$, the minimum restoration witnesses are exactly the bases of M_D. In particular, they satisfy basis exchange.*

Proof. The pivot must be restored precisely when $p \in D$, so it is present in every minimum witness in that case. The retained support meets $4 - t(D)$ direction blocks. Reaching two directions therefore requires one restored label from each of exactly $r(D)$ distinct fully deleted blocks. Choosing two labels from the same block creates no additional met direction and cannot occur in a minimum witness. Conversely, the pivot choice together with any $r(D)$ such distinct-block choices is a full repair. These choices are precisely the bases of the displayed direct sum. □

The independent checker does not infer this result from the formula. It reconstructs the C75 point-set closure table, tests restoration subsets in increasing cardinality for every D, compares the resulting family with $\mathcal{B}(M_D)$, and checks all 198912 ordered exchange obligations.

3 Ten templates and five exchange graphs

Join two bases when their symmetric difference has size two. The exact template atlas is

ρ	W	$\#D$	exchange graph
0	1	30400	K_1
1	1	30400	K_1
1	4	1984	K_4
1	7	192	K_7
1	8	128	K_8
2	4	1984	K_4
2	7	192	K_7
2	8	128	K_8
2	25	64	$L(K_{1,1,2,5})$
3	25	64	$L(K_{1,1,2,5})$

For $t \leq 2$ the direction rank is zero, giving K_1. For $t = 3$ the direction rank is one, so all W singleton choices exchange with one another, giving K_W for $W = 4, 7, 8$. For $t = 4$, the rank-two bases are the edges of the complete multipartite graph $K_{1,1,2,5}$, and exchange adjacency is edge incidence; hence the basis graph is its line graph.

Aggregating the two pivot states gives the finite graph atlas

type	$\#D$	$\|V\|$	$\|E\|$	diam	degree multiplicities
K_1	60800	1	0	0	0^1
K_4	3968	4	6	1	3^4
K_7	384	7	21	1	6^7
K_8	256	8	28	1	7^8
$L(K_{1,1,2,5})$	128	25	128	2	$9^{10}, 10^{10}, 13^4, 14^1$

4 The all-deleted atlas

When $D = L$, every basis is

$$\{S_9, x, y\}, \qquad x \in B_i, \quad y \in B_j, \quad i \neq j.$$

There are

$$1 \cdot 1 + 1 \cdot 2 + 1 \cdot 5 + 1 \cdot 2 + 1 \cdot 5 + 2 \cdot 5 = 25$$

such bases. Expanding the seven C76 full-core-minimal representatives under the effective label generators gives the same 25 masks, not merely the same cardinality. The relevant action has order 1920; it is distinct from the order-11520 ambient lift bound in C75.

For an edge joining parts of sizes a and b in $K_{1,1,2,5}$, the corresponding line-graph degree is $16 - a - b$. This gives the exact spectrum

$$\{9^{10}, 10^{10}, 13^4, 14^1\}.$$

The graph has 128 edges, radius and diameter two, and 172 unordered vertex pairs at distance two (the other 128 pairs are adjacent).

5 Audit boundary

The producer byte-binds C75, C76, and C79 evidence and manifests. A separate SymPy block polynomial recovers all ten support counts, while an exact 25×25 adjacency-matrix calculation recovers the line-graph invariants. Clean replay is byte-stable and all 18 hostile semantic mutations are rejected. The canonical evidence SHA-256 is

`9c3b20c703b680a391ad1834c0f55cabaf27bfed14cee2099b0c3afa1eb259ca`

No arithmetic/local data, Euler factor, root number, automorphy statement, full Burnside ring or table of marks, or Hilbert–Pólya operator is claimed.